MELVIN TABITHA OPONDO

Full-Stack Software Developer

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PROFESSIONAL SUMMARY

I am a full-stack Software Developer with hands on experience building responsive web applications using React.js, Django, and Flask. I design RESTful APIs, manage relational and non relational databases, and deploy scalable systems. I work across the full development lifecycle, from architecture and backend logic to frontend interfaces and API integration.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, HTML, CSS
Frontend: React.js, React Native, TailwindCSS, Bootstrap, Vite, React Hook Form, React Router
Backend & Frameworks: Django, Django REST Framework, Flask, FastAPI
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis
Tools & DevOps: Git, GitHub, Docker, Kubernetes (learning), Postman, Ngrok, Vercel, Render, Neon, Oracle Cloud, Netlify, Firebase
APIs & Integrations: RESTful API development, Safaricom Daraja (M-Pesa), Google Maps API, OpenWeather, JWT Authentication
Testing: Pytest, Django Test Framework, React Testing Library, Jest
Other: Technical SEO, Canva, Adobe Photoshop

EXPERIENCE

Computer Science Academy Africa (CSA)
Web Developer, Website Maintenance & Technical SEO | csaafrica.org

- I designed, developed, and deployed the live website for Computer Science Academy Africa, configuring its production hosting on the organisation's accounts and implementing the technical SEO fundamentals that keep the organisation discoverable to the students and partners it exists to reach.
- I take responsibility for a real external client whose users depend on the site being up and findable, not a weekend project. I handle deployment, monitoring, and ongoing content and performance updates.
- I apply technical SEO practices (clean site structure, crawlable markup, descriptive metadata, performance oriented page delivery) so the organisation reaches its audience through organic search rather than paid acquisition alone.

ACSIS, African Caribbean Sustainability & Investment Summit
Web and Graphic Designer

- I designed graphics and visual materials that supported the organisation's communications, branding, and visibility.
- I worked collaboratively with a team of designers, coordinating on shared design work so that each piece aligned with the organisation's branding requirements.
- I contributed to keeping the organisation's materials visually consistent and to improving the presentation of its communications.

FEATURED PROJECTS

- **Anga.** Django REST and React weather platform for 99 Kenyan locations, built around a 1,000 request per month API limit: two-tier Redis caching with stale fallback, request coalescing with atomic Redis locks (30 concurrent requests, 1 upstream call), a 429 circuit breaker, a quota reserve, and per-client throttling. PostgreSQL gazetteer for search, 222 automated tests. Sole developer. (Live | Code)
- **StudyBuddy.** Django REST and React 19 study assistant that verifies every AI-generated definition and quiz question against the student's own notes before display. Gemini model fallback and SHA-256 result caching on a quota of about 20 requests a day; custom JWT auth with refresh-token rotation, Argon2 hashing, and file-signature upload checks. Sole developer. (Live | Code)
- **CallShield.** AI-driven real time scam call detection, aimed at giving users a signal before they engage with a scam rather than after they lose money. Django and REST backend, React Native frontend, PostgreSQL. Technical ownership plus Project Manager role.

- **SmartHunt.** Student housing platform with Safaricom Daraja (M-Pesa) payment integration, meeting Kenyan users on the mobile money infrastructure they already use. Google Maps powered location search, Django and FastAPI backend, React frontend.
- **UjamaaFlow.** Flask and SQLAlchemy donation platform with three-role accounts (donor, recipient, administrator), photo listings with category and location filtering, admin matching of donations to requests, and a community-report moderation queue. (Live | Code)
- **Master Photography Kenya.** Ten-page studio website in HTML, CSS, and vanilla JavaScript with no third-party requests, generated by a custom Python build tool and gated by an audit script that fails on broken links, missing alt text, or invalid JSON-LD. WhatsApp enquiry flow with Kenyan phone number validation. (Live | Code)
- **Developer Portfolio.** Static site in HTML, CSS, and vanilla JavaScript with no framework or build step: an interactive terminal, light and dark themes, accessible defaults, and SEO foundations. Deployed on Vercel. (Live)

EDUCATION & TRAINING

Meru University of Science and Technology. Bachelor of Science in Computer Science
Computer Science Academy (CSA). Software Engineering
Power Learn Project (PLP). Software Engineering
Africa Agility Foundation. Frontend Web Development
Workmates Technology. UI/UX Design
ALX, AI Career Essentials Program. AI tools integration and ethical AI applications in industry workflows
Kolanya Girls' School. KCSE | 2019 to 2022

LEADERSHIP & VOLUNTEER

Community Outreach Specialist. WISSA, Women in Sustainable STEM Africa. I supported outreach programmes aimed at empowering women and girls in STEM fields across Africa.
Web & Graphic Designer (Volunteer). ACSIS, African Caribbean Sustainability & Investment Summit. I designed digital graphics and promotional materials using Adobe Photoshop and Canva.
Mentor & STEM Advocate. Kolanya Girls' Secondary School. I mentored students in STEM career pathways, leadership development, and facilitated coding workshops.
Social Media Manager. Blockchain Community, Meru University Science Innovators Club. I managed and grew the community's social media presence, creating technical and educational content on blockchain and emerging technologies.

SOFT SKILLS

Leadership & Responsibility • Problem Solving & Analytical Thinking • Team Collaboration • Adaptability & Continuous Learning • Effective Communication

LANGUAGES

English, Fluent | Kiswahili, Fluent | French, Intermediate